

Zelin Ye

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EDUCATION

Carnegie Mellon University, U.S. *M.S. in Electrical and Computer Engineering* | GPA: 3.93. May 2022

Beijing University of Technology, China. *B.S. in Electronic Information Engineering* | GPA: 3.86. July 2020

SKILLS

Programming: (Proficient) Python, C/C++, X86 Assembly, MATLAB, JavaScript, HTML

Software and platform: PyTorch, TensorFlow, Sklearn, NumPy, GDB, LaTeX, Tkinter, Multisim, Linux, AWS, ROS, STM32

Coursework: Computer Systems, Computer Vision, Machine Learning, Deep Learning, Visual Learning, Convex Optimization.

WORKING EXPERIENCE

Carnegie Mellon CyLab

Pittsburgh, PA

Research Assistant

Nov 2021 – May 2022

- Deployed MMDet-based product detection pipeline to NVIDIA Jetson AGX Xavier. Optimized runtime from 8 seconds to less than 0.15 seconds per image through thresholding NMS and code optimization via Python and CUDA.
- Converted the Swin-Transformer-based feature extraction and matching models into ONNX format and deployed with TensorRT using PyTorch. Formed Multiview feature combining pipeline for product label searching.

Fujian (Quanzhou)-HIT Research Institute of Engineer and Technology

Fujian, China

Hardware Engineer Intern

July 2020 – Nov 2020

- Controlled poses of robotic arms using one of the architectural design patterns (publish-subscribe pattern, pub/sub) communication protocol in Robot Operating System (ROS) with C++.
- Calibrated hand-eye relative pose by computing transformation matrix from coordinates of the camera to robotic arm base.
- Validated calibration accuracy through object catching using UR10 robotic arm and Hikvision camera.

RESEARCH EXPERIENCE

Object detection based on Bayesian multi-modal fusion - ARGO AI | NAVLAB

Jun.21 – Jun.22

- Designed the hardware structure of beam splitter divided RGB and thermal cameras, and collected a spatial and temporal aligned multimodal dataset through video rectification and an external trigger on NAVLAB11 in cooperation with Freiburg.
- Proposed a Probabilistic Ensemble algorithm as the first author that obtained 13% improvement over state-of-the-art method on FLIR and KAIST dataset.

Question-Image Co-Attention for Visual Question Answering

Aug.21 – Dec.21

- Generated VQA data loader, created word-level question embedding, sentence-level answer embedding, and extracted visual features with GoogLeNet and ResNet. Implemented image caching methods for large batch size training and testing.
- Applied alternating co-attention mechanism to combinedly attend to image and 3-level hierarchical question feature. Trained model with weight clip, different learning rates and RMSProp, achieved more than 50% answer predict accuracy on VQA1.0.

Attention-based End-to-End Speech-to-Text Deep Neural Network

Aug.21 – Dec.21

- Implemented Listen, Attend, Spell character-based model to predict scripts by utterances with variable duration.
- Improved model with weight tying, a more complex encoder, and an attention layer to achieve an average Lev distance of 18.

Foundations of Computer Systems

Jan.21 – May.21

- Cache Lab: Simulated the hit/miss/evict behavior of cache memory by LRU and optimized traverse order to improve the time and space locality by blocked matrix multiplication utilizing cache space only containing 8 bits for temp constant data.
- Shell Lab: Created a tiny shell executing commands in a multi-process C program, processed signals to avoid race conditions.
- Malloc Lab: Implemented the malloc and free function using segregated linked lists.
- Proxy Lab: Constructed a proxy-based on socket interfaces with C. Dealt with multiple concurrent connections using multi-thread programming. Cached recently used web content.

Augmented Reality Application with Planar Homographies

Jan.21 – May.21

- Developed an **Augmented Reality** application. Overlaid warped user-input movie to the recorded video of a moving target template, displayed augmented output on a webpage and a comparable Tkinter GUI interface.
- Dynamically recognized matching points using RANSAC for autonomous inter-frames homography estimation and designed constraints to suppress potential drifting.

Loudness Compensation Algorithm in Digital Hearing Aids

Jan.20 – May.20

- Devised a multi-channel loudness compensation algorithm to suppress residue noise based on noise tracking, gained 12% improvement in intelligibility, 11% in clarity, and ensured user comfort with MATLAB.

PUBLICATIONS & PATENTS

- Zelin Ye***, Yi-Ting Chen*, Jinghao Shi*, Christoph Mertz, Shu Kong*, Deva Ramanan*. "Multimodal Object Detection via Probabilistic Ensembling." ECCV 2022 Accept-Oral. Link: <https://arxiv.org/abs/2104.02904>.
- Ruwei Li, Kaixuan Dai, **Zelin Ye**, and Yongya Zhang. "Multi-channel adaptive loudness compensation algorithm based on noise tracking in digital hearing aids." Speech Communication 130 (2021): 64-76.
- Software copyright: "Codelink" Blockchain Traceability Platform Management System NO. 2019SR085360